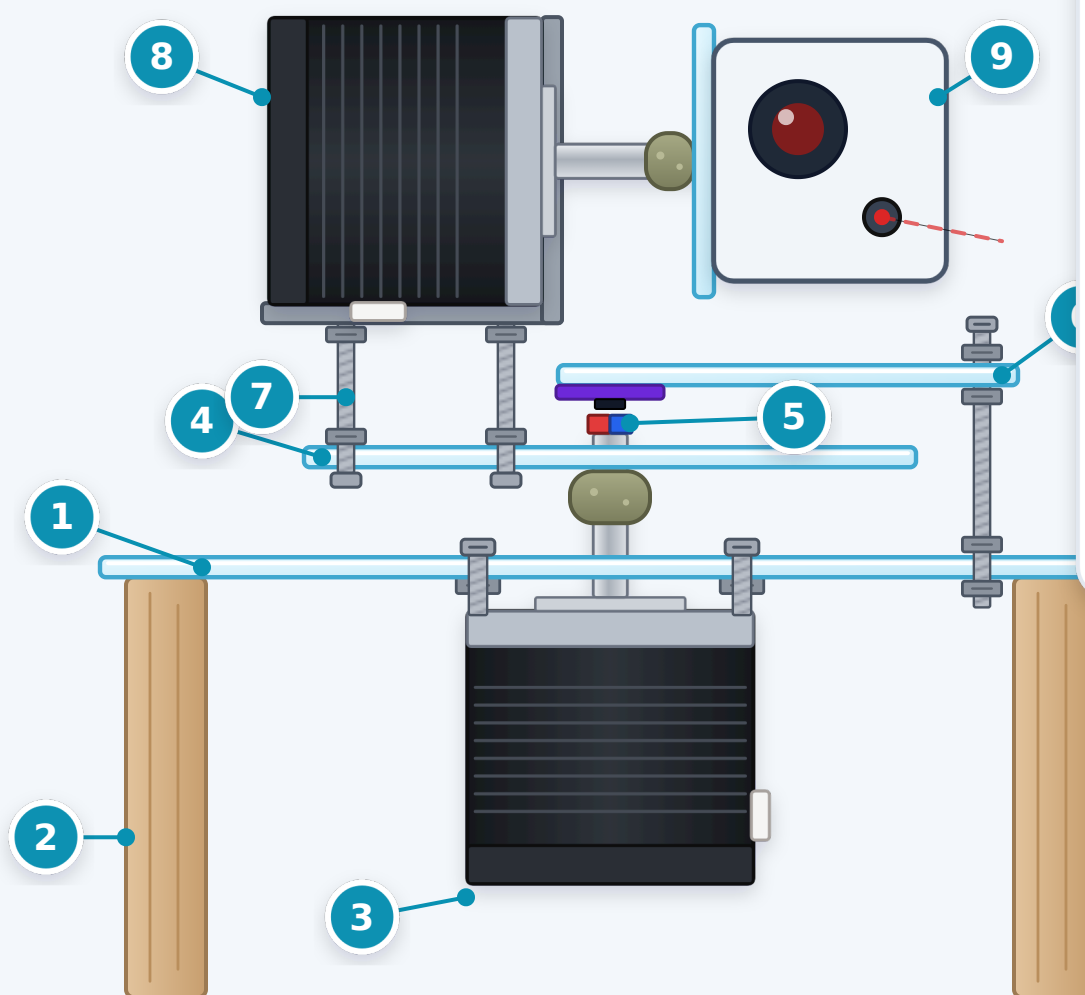




This is what you are building

Side view of the finished MK2

bottom motor turns left-right · top motor nods up-down



- 1 Base plate (acrylic)
- 2 Legs (wood blocks)
- 3 Bottom motor (pan)
- 4 Big disc (M-Seal)
- 5 Magnet on shaft tip
- 6 Sensor arm + AS5600
- 7 Stilts (long screws)
- 8 Bracket + top motor
- 9 Small disc + head

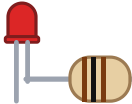
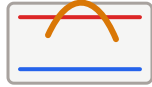
- 1 Two motors, stacked. The **bottom** one turns the whole thing left-right (pan).
- 2 The **top** one, mounted sideways on a bracket, nods the head up-down (tilt).
- 3 The camera and the laser sit together in the **head**, so the camera always sees its own laser dot.
- 4 Everything is flat acrylic pieces, screws and M-Seal. **No 3D printing, no coupler, no belt.**
- 5 Two more parts sit apart from the rig: the blinking **beacon** (the target) and the steady **decoy** (the trick light).



Tonight's goal: build this, wire it, and watch it lock onto the blinking beacon on the website.



What you have, and the only 3 things to get

 2 motors (NEMA17) ✓	 2 motor drivers ✓	 2 Arduino Nano ✓	 2 sensors + magnets ✓
 Multiplexer board ✓	 Laser (KY-008) ✓	 USB webcam (arriving) ✓	 LEDs + resistors ✓
 Acrylic sheets ✓	 Screws + nuts ✓	 12 V adapter + DC socket ✓	 Breadboard + wires ✓
 Ping-pong ball + AA holder ✓	 Glue gun + solder ✓	 Multimeter ✓	 Cable ties ✓

GET FROM ANY HARDWARE / GENERAL SHOP (if not at home)

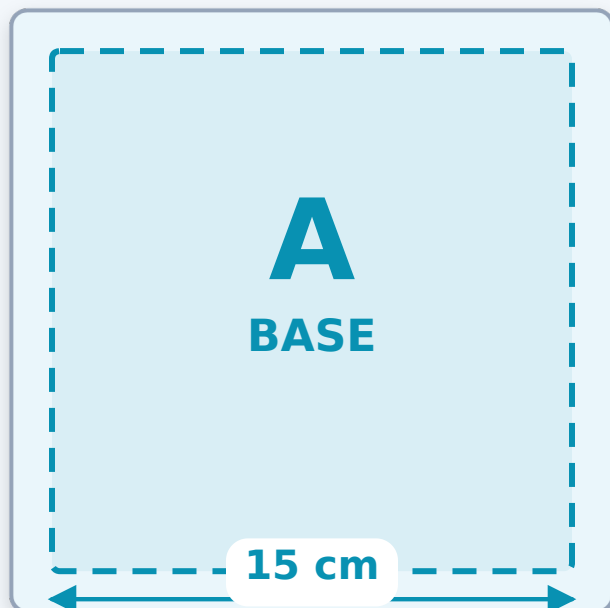
 M-Seal or Araldite ₹50-100 · glues the discs	 3 AA cells ₹30 · the decoy	 Fevikwik ₹10-20 · the magnet
--	--	--

- 1 Everything with a green tick you **already have**. Count each item out onto the table before you start.
- 2 No webcam yet? Use the laptop's own camera for every step tonight — swap it in later, nothing else changes.
- 3 The orange box lists the only 3 things to buy, and any hardware, paint or general store has them.
- 4 **M-Seal (or Araldite) is the one thing you can't skip** — it is what glues both discs onto the shafts.
- 5 No coupler, no red acrylic, no belt, no 3D-printed part is used anywhere in this guide.
- 6 Keep the yellow bellows coupling you bought as a spare — this build does not need it.

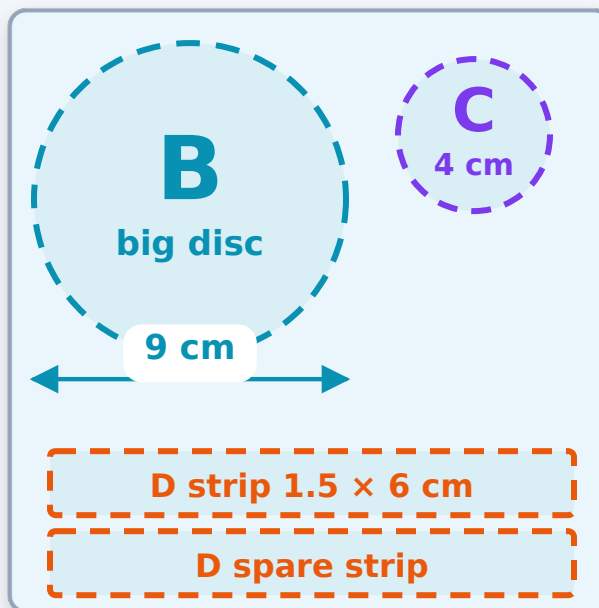
✓ **Nothing else to buy. Everything in this guide uses only these parts.**

Cut 4 pieces from the acrylic

SHEET 1



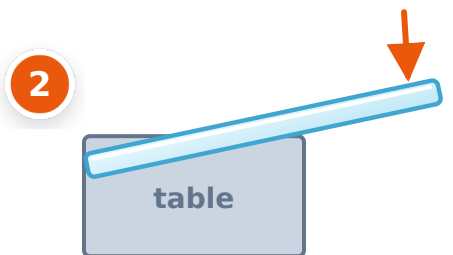
SHEET 2



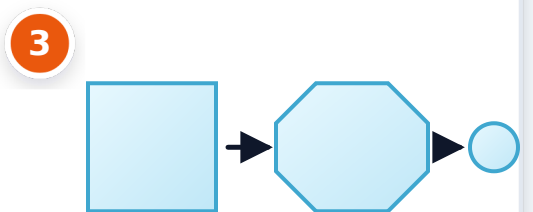
HOW TO CUT



Scratch the line 8-10 times with a cutter, along a ruler



Line on the table edge. Press down hard: it snaps clean.



Square → cut corners → sand it round.

YOU NEED

- | | |
|-----------------------|-------------------|
| A Base | 15 × 15 cm square |
| B Big disc | 9 cm circle |
| C Small disc | 4 cm circle |
| D Sensor strip | 1.5 × 6 cm |

Keep the paper film on while cutting — no scratches.

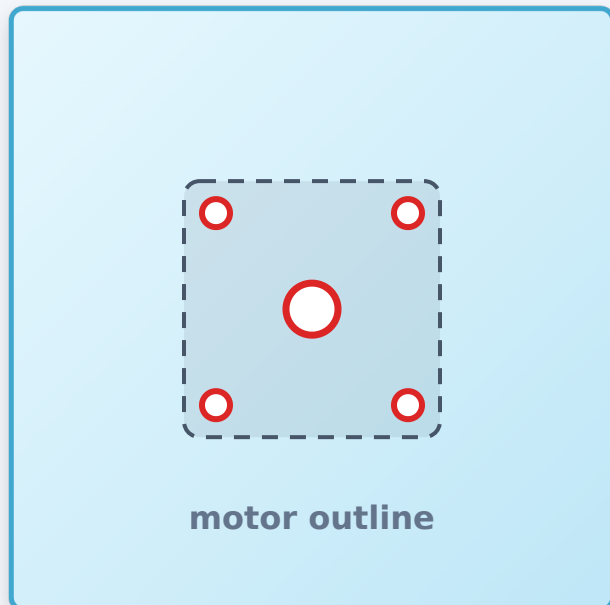
Tools / parts for this page: Acrylic sheets, steel ruler, cutter/blade, marker pen, a round cup or lid, sandpaper.

- 1 Keep the protective paper film ON. Mark all 4 shapes on the sheets first, with the ruler and marker.
- 2 **Straight edges** (base, strip): scratch the line 8-10 times with a cutter along the steel ruler.
- 3 Line the scratch up with the edge of a table, then press down hard — it snaps clean along the line.
- 4 **Circles** (both discs): cut a rough square around the circle, then snip off the 4 corners.
- 5 Sand the disc's corners round with sandpaper. It does not need to be a perfect circle.
- 6 Peel the paper film off only at the very end, after every piece is cut.

4 pieces ready: base, big disc, small disc, one sensor strip (plus a spare).

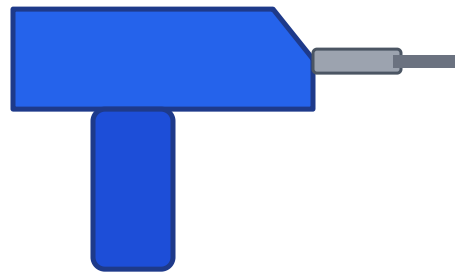
Make the holes

MARK THEM



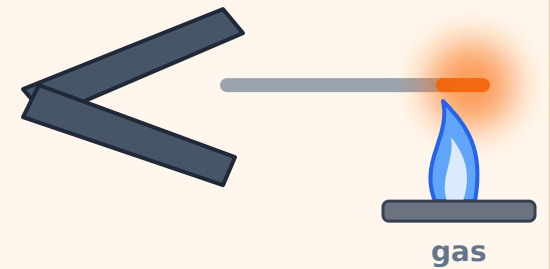
Motor face-down in the middle. Dot through its 4 holes, and the middle.

WITH A DRILL



Small tape on the spot so the bit does not slide. Go SLOW, no pressure.

NO DRILL? HOT NAIL



Hold a nail with pliers. Heat the tip on the gas 20 seconds. Push it through the dot.

Never touch the tip!

WHICH HOLES

- A Base** 4 small (screws) + 1 big middle hole (8 mm, the shaft must NOT touch)
- B Big disc** 1 middle hole = exactly the shaft size (tight) + 3 small holes on ONE side
- C Small disc** 1 middle hole = exactly the shaft size (tight) + 2 small holes
- D Strip** 1 small hole at one end

Tools / parts for this page: The motor itself (as a stencil), marker pen, drill with a small bit OR a nail + pliers + gas stove, screwdriver, sandpaper.

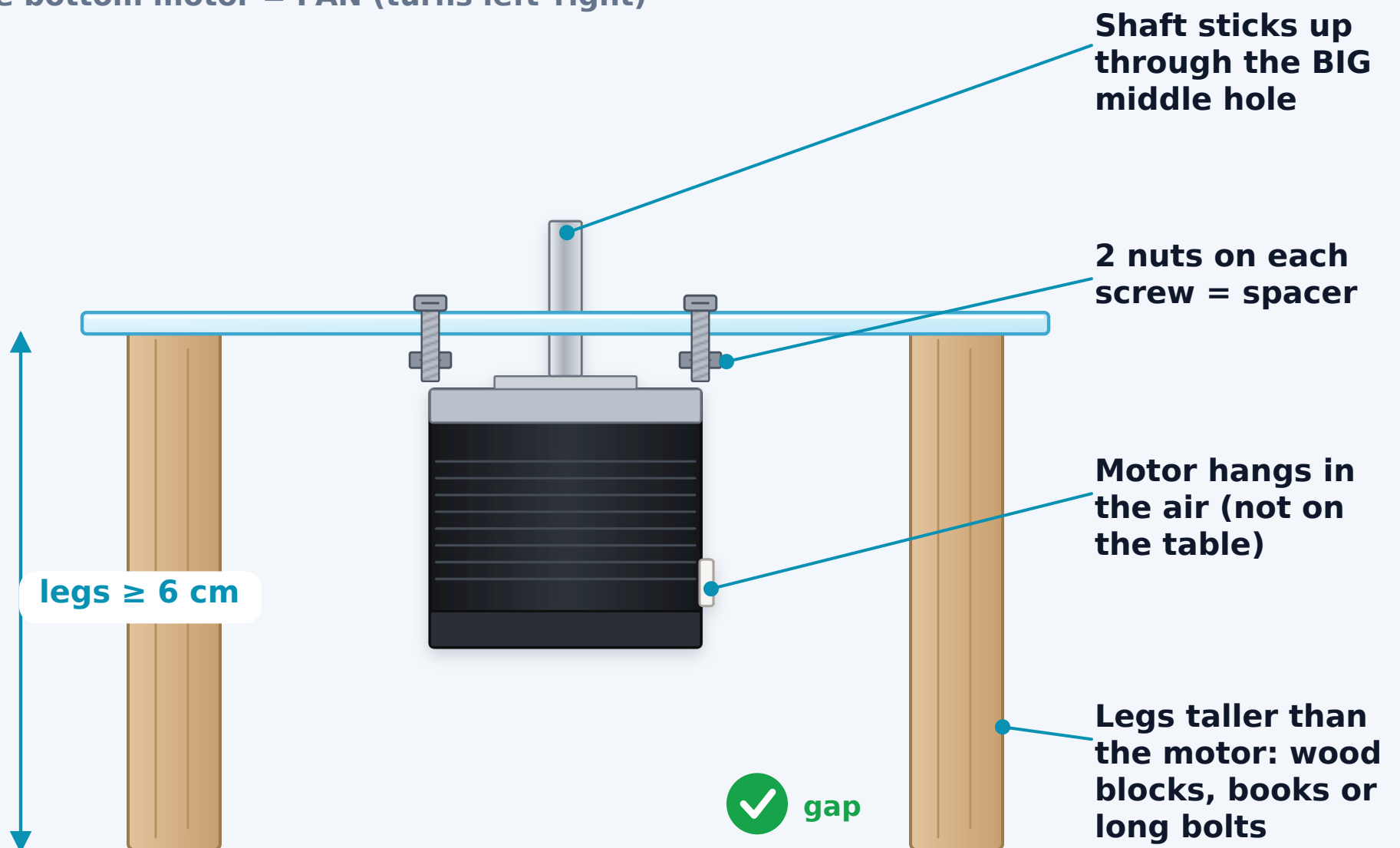
- 1 Lay the motor face-down on each piece and dot through its 4 screw holes and the middle — this makes every piece match.
- 2 **With a drill:** put a small piece of tape on the dot so the bit doesn't slide. Drill slowly, with no pressure.
- 3 **No drill:** hold a nail with pliers, heat the tip on the gas for 20 seconds, then push it straight through the dot. Never touch the tip.
- 4 On the **base**: make the middle hole big (about 8 mm) — the shaft must spin freely inside it, touching nothing.
- 5 On both **discs**: start the middle hole small, then widen it slowly by twisting a screwdriver in it, until it is tight.
- 6 Test-fit: push a disc onto the motor shaft. It should go on snug — not loose, not forced.

Push a disc on the motor shaft: it goes on snug, not loose and not forced.

Base, legs and the bottom motor

Side view

the bottom motor = PAN (turns left-right)



Tools / parts for this page: Bottom NEMA17 motor, base plate, 4× M3×12-16mm screws, 8+ M3 nuts, screwdriver, 4 legs (wood blocks, books or long bolts, ≥ 6 cm).

- 1 Hold the motor under the base, and push its shaft **up** through the big middle hole.
- 2 Thread **2 nuts onto each of the 4 screws** first — these act as spacers between the base and the motor.
- 3 Push the 4 screws down from the top, through the base, into the motor's 4 corner holes, and tighten by hand.
- 4 Stand the base on 4 legs, one at each corner — the legs must be **taller than the motor** so it hangs free.
- 5 Check the base sits flat on the table with no wobble.

✓ The base sits flat and does not rock. The motor does not touch the table.

Glue the big disc onto the shaft

Close-up, side view

the disc is glued straight onto the metal shaft

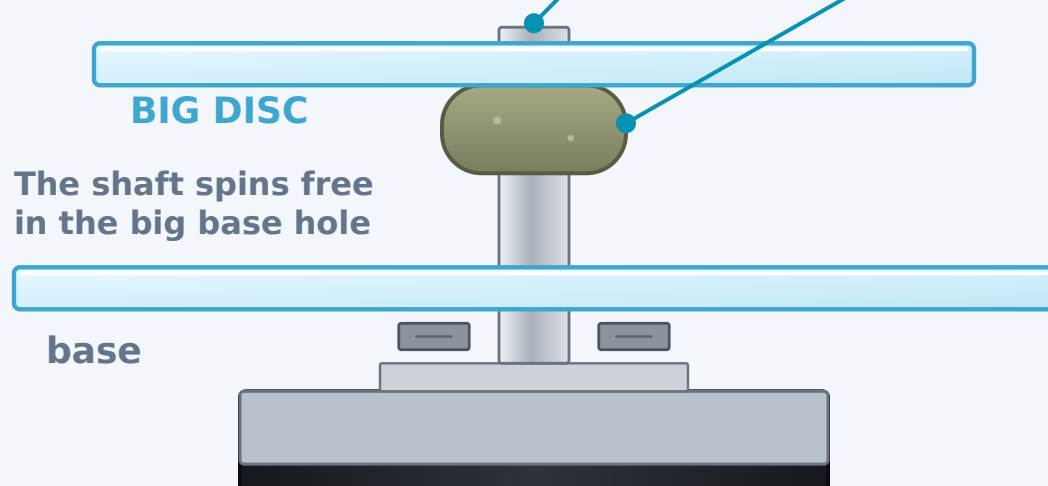
KEEP IT FLAT



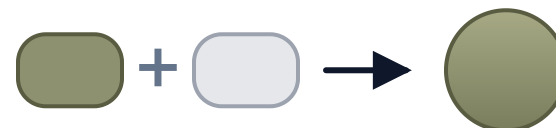
Phone level app on the disc: bubble in the middle = flat.

Shaft tip sticks out about 2 mm — keep it **CLEAN** (magnet goes here next)

Fat M-Seal ring **UNDER** the disc, all round the shaft



MIX M-SEAL



Equal bits of both colours. Roll till it is **ONE** colour.

Hard in about 1 hour.

Tools / parts for this page: M-Seal or Araldite, the big disc, phone with a level app.

- 1 Break off equal-sized bits of the M-Seal's two colours and roll them together until it is **one solid colour** (about a minute).
- 2 Push the big disc onto the motor shaft until only about **2 mm** of the shaft tip sticks out on top.
- 3 Press a fat ring of the mixed M-Seal **under** the disc, all the way round the shaft.
- 4 Keep the very **top of the shaft clean and dry** — no putty up there, the magnet goes on that spot next.
- 5 Rest a phone with a level app on the disc — the bubble should sit in the middle, meaning it is flat.
- 6 **Don't touch or turn it for 1 hour** while the M-Seal hardens.

After 1 hour: spin it by hand. It turns flat, no wobble, and the disc never slips on the shaft.

Magnet and the sensor arm



Magnet test: N and S
on the **SIDES** (half red,
half blue) = correct

Close-up, side view

Magnet glued on
the **TIP** of the shaft
(Fevikwik)

Sensor chip faces
DOWN, exactly over
the magnet

Long screw = post.
It stands on the base
and **NEVER** turns.

Nuts above and
below the strip set
the height

2 credit cards fill the gap —
slide them out after

base

 **Tools / parts for this page:** Diametric magnet, Fevikwik (superglue), AS5600 sensor, acrylic strip (D), M3×40mm screw + nuts, 2 credit cards.

- 1 Test the magnet first: held flat, it should show **red on one side, blue on the other** — that is the correct diametric type.
- 2 Put one drop of Fevikwik on the very tip of the (now hardened) shaft, and press the magnet on flat.
- 3 Stand the M3×40 mm screw upright on the base, 4–5 cm off-centre, and lock it in place with a nut.
- 4 Slide the acrylic strip onto the screw so its far end reaches out over the magnet.
- 5 Slip 2 credit cards between the magnet and the strip, as a spacer, then set 2 more nuts to hold the strip at that height.
- 6 Stick the AS5600 sensor onto the strip's end, **chip facing down**, resting against the 2 cards.
- 7 Slide the 2 cards back out — the sensor now hovers about 1.5 mm above the magnet, touching nothing.

 **Spin the disc: the strip and the sensor never move and never touch anything.**

Stilts, bracket and the top motor

Side view of the top half

Top motor = TILT
(nods up-down)

L-bracket

3 long screws =
stilts, on the side
AWAY from the
sensor arm

shaft points
SIDEWAYS

gap

The bracket must pass
OVER the sensor arm with
a gap. Turn the nuts on
the stilts to raise it.

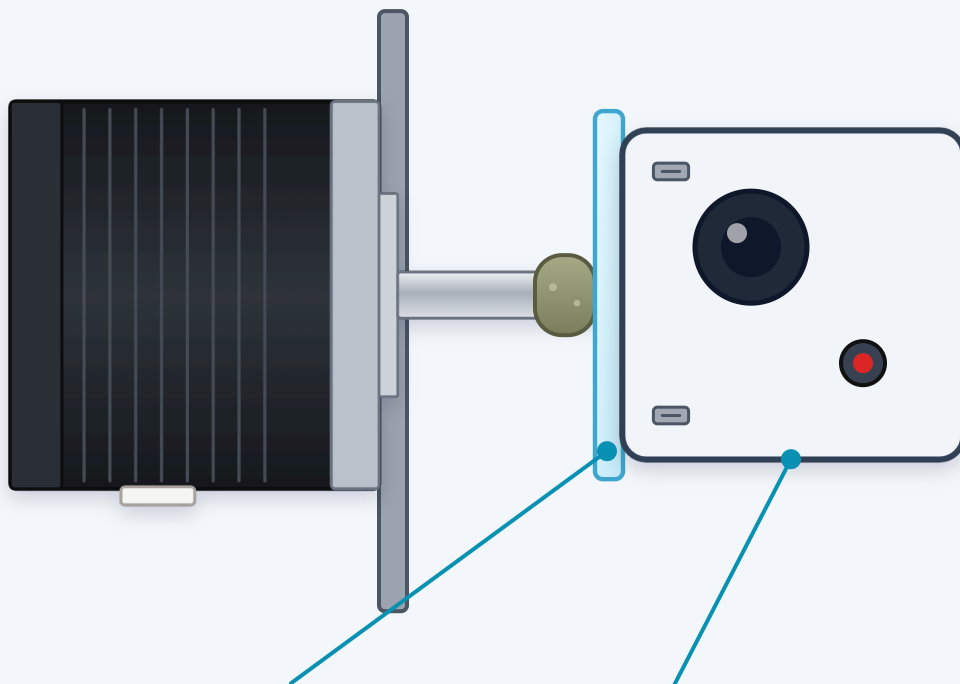
 **Tools / parts for this page:** L-bracket, top NEMA17 motor, 3× M3×40mm screws (stilts), M3 nuts, 4× M3×6mm screws, screwdriver.

- 1 Push 3 long screws up through the big disc's small holes, on the side **away from** the sensor arm.
- 2 Lock each stilt screw from below with a nut, close to the disc.
- 3 Rest the L-bracket's flat foot on top of the 3 stilts, and add a nut above and below it to hold its height.
- 4 Turn the disc a quarter turn left and right by hand — the bracket must clear the sensor arm. Raise the stilt nuts if it doesn't.
- 5 Bolt the top motor to the bracket's standing part with 4 screws, **shaft pointing sideways**.

 Turn the big disc a quarter turn left and right by hand: the bracket never hits the sensor arm or its screw.

Small disc and the head

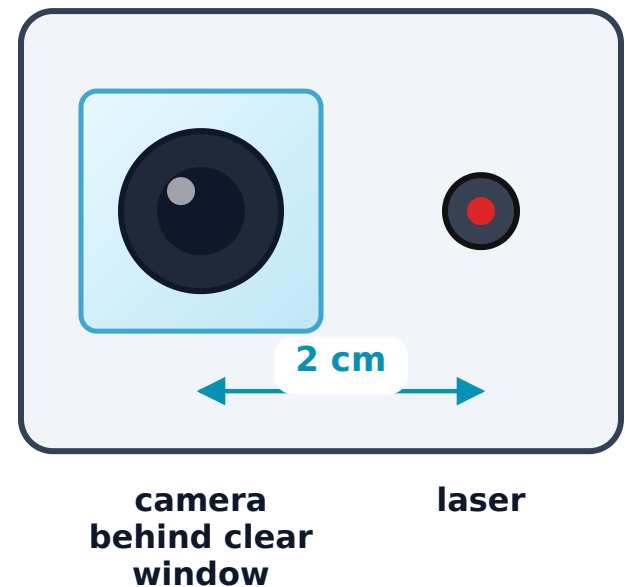
Side view of the head



Small disc glued on the top shaft (same M-Seal trick)

Head box screwed **FLAT** on the small disc, centred → it balances

FRONT OF THE HEAD



Both look the **SAME** way. Hot-glue them.

The 2nd sensor is **OPTIONAL**

The top motor works without it. Skip it tonight. The code now works with just the bottom sensor.

Leave the cables loose, tie them with cable ties: slack at every joint.

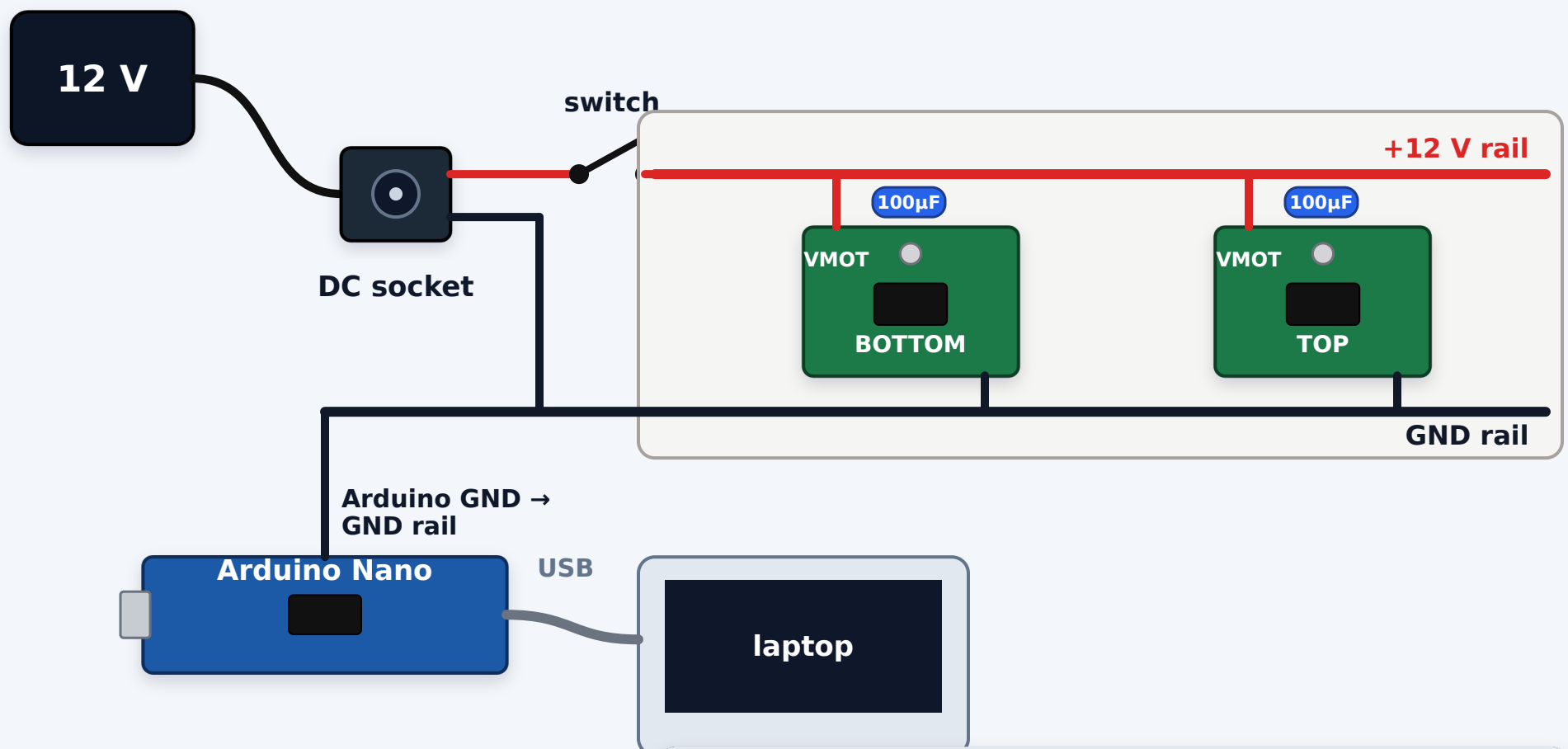
 **Tools / parts for this page:** M-Seal, small disc, head box, screws, hot glue gun, webcam, laser module.

- 1 Glue the small disc onto the top motor's shaft the same way as before: M-Seal ring under it, 2 mm sticking out.
- 2 Wait 1 hour, untouched, for this M-Seal to harden too.
- 3 Screw the head box **flat onto the small disc**, centred — centred means balanced, so the motor doesn't strain.
- 4 Hot-glue the webcam inside, facing forward, right behind the clear window.
- 5 Hot-glue the laser 2 cm beside the camera, pointing the exact same way.
- 6 Leave the camera and laser cables loose with slack at every joint, so they don't pull when the head turns.
- 7 Skip the 2nd sensor tonight — it is optional, and the code works fine with just the bottom sensor.

 On a wall 2 m away the laser dot lands near the middle of the camera picture.

Power: 12 V for the motors only

Two powers, ONE ground



Find + on the socket:
adapter in, multimeter
on DC V. The 2 pins
that read 12 V: red
probe = +. Mark it.

- 12 V goes **ONLY** to the motor drivers (VMOT)
- Never pull a motor wire while the switch is ON
- Capacitor: long leg to +12 V, stripe to GND
- Arduino GND to the GND rail — one shared ground

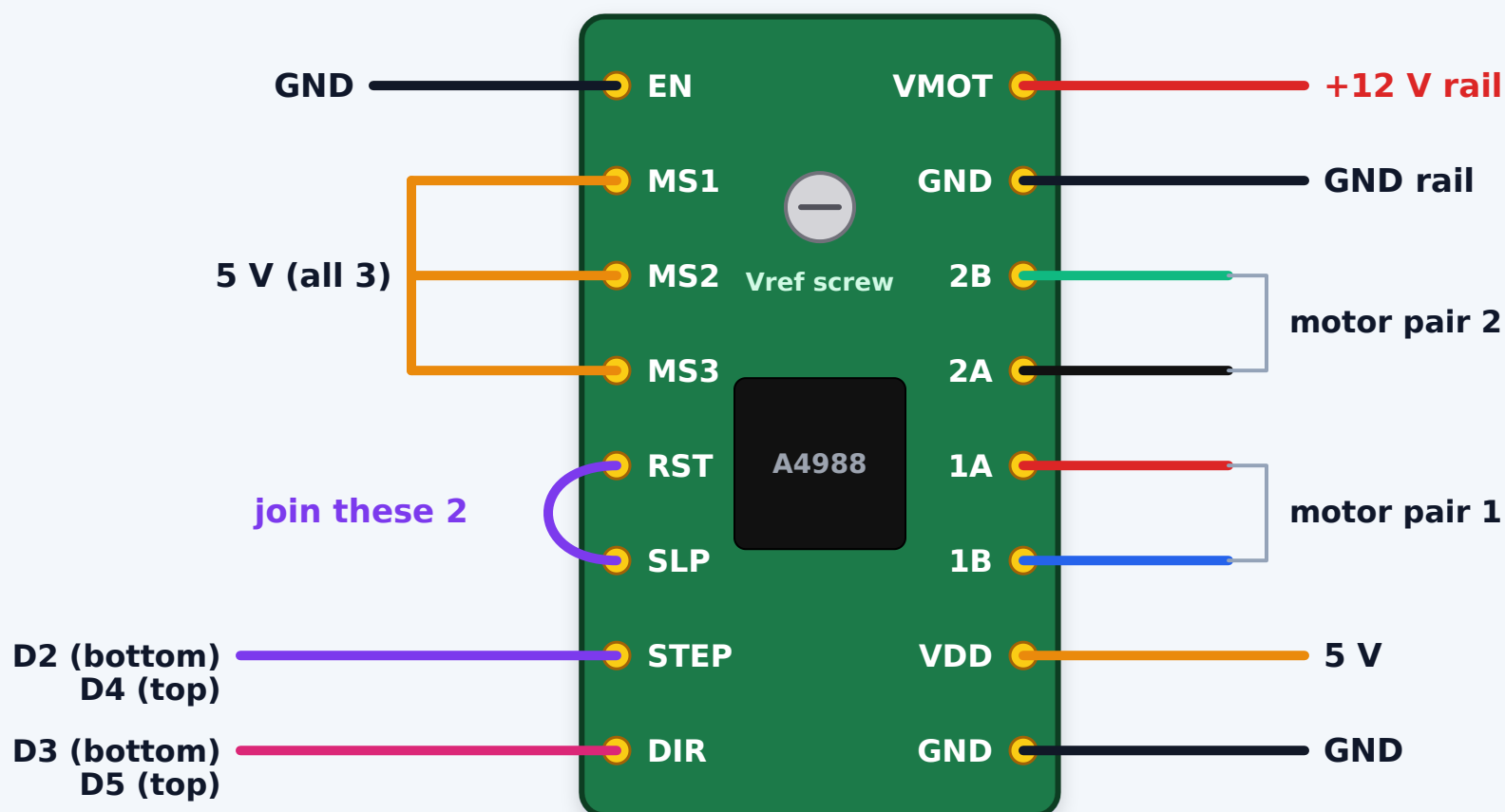
Tools / parts for this page: 12 V adapter, DC socket, switch, breadboard, 2× 100 µF capacitors, multimeter, jumper wires.

- 1 Find the socket's + pin: plug in the adapter, set the multimeter to DC V, touch its 2 pins — the **red probe reading positive** is +. Mark it.
- 2 Wire: **12 V adapter** → DC socket → switch → the **red (+12V)** rail on the breadboard.
- 3 Wire the socket's – pin straight to the **blue (GND)** rail.
- 4 On each driver: VMOT → red rail, GND → blue rail, with a 100 µF capacitor across them (long leg to +, stripe to GND).
- 5 Power the Arduino separately, from the **laptop's USB** — never from the 12 V rail.
- 6 Join one wire from the Arduino's GND pin to the GND rail — this is the single shared ground for everything.
- 7 Switch on with nothing else connected yet: the multimeter should read about 12 V on the red rail.

✓ **Switch ON (nothing else plugged in yet): the red rail reads about 12 V on the multimeter.**

Wire a motor driver (do it twice)

One driver · wire it the same way twice



Find the 2 pairs

Multimeter on Ω : the 2 wires that read 1-4 Ω = one pair. Usually black+green and red+blue. Never mix the pairs.

Vref (USB only, 12 V OFF)

Black probe GND, red probe on the little screw. Turn gently to 0.55 V. Do both drivers.

Tools / parts for this page: 2× A4988 driver, multimeter, small screwdriver (for Vref), jumper wires.

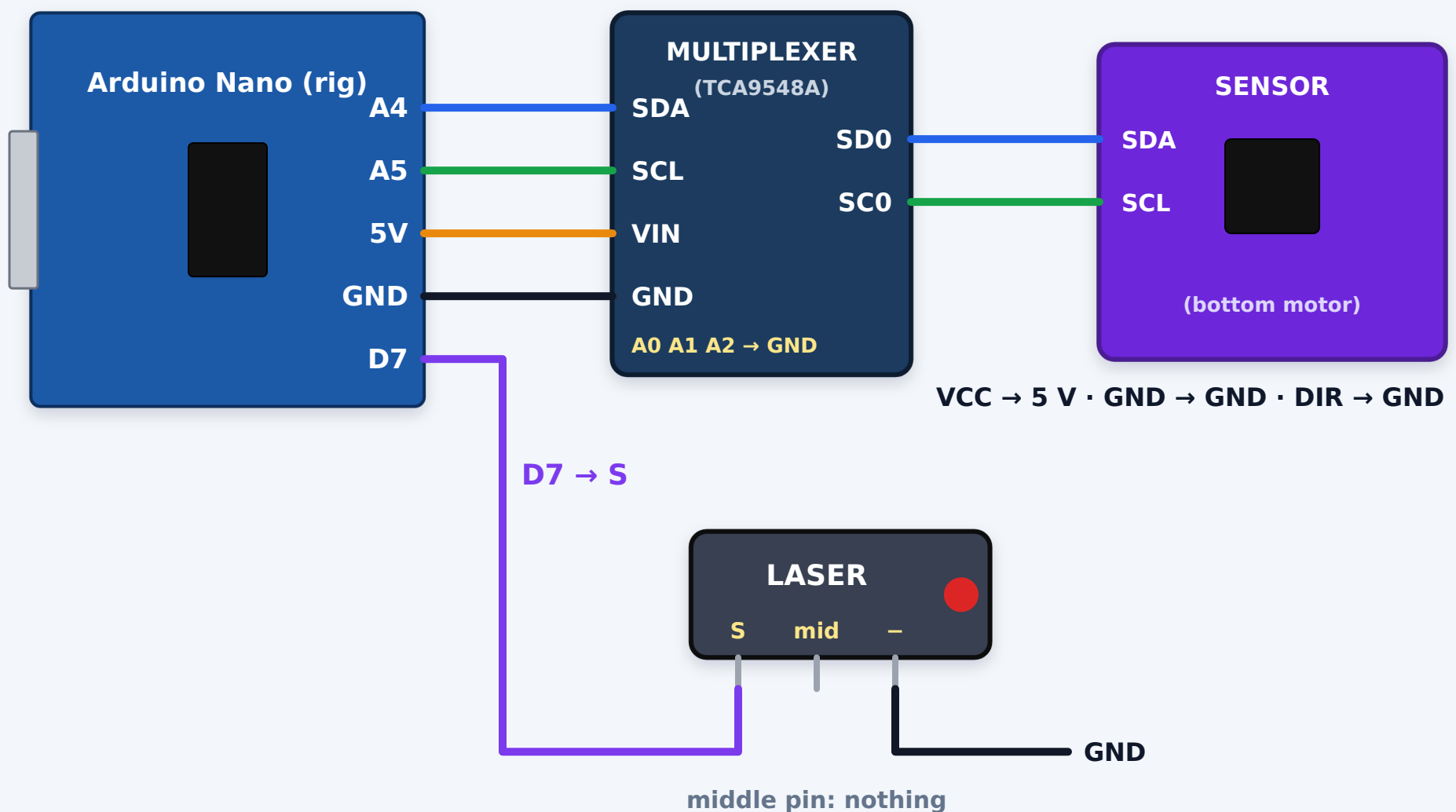
- 1 Find the motor's 2 coil pairs first: multimeter on Ω , the 2 wires that read 1-4 Ω are one pair. Never mix the pairs.
- 2 Small jumpers on the driver: join **RST**↔**SLP** together, **MS1 MS2 MS3** to 5 V, and **EN** to GND.
- 3 Wire **STEP** / **DIR** to the Arduino: bottom motor to **D2** / **D3**, top motor to **D4** / **D5**.
- 4 Wire one coil pair to **1A-1B**, and the other pair to **2A-2B**.
- 5 With USB power only and the 12 V switch OFF, turn the small screw gently until **Vref reads 0.55 V**. Do this on both drivers.
- 6 Switch the 12 V on: both motor shafts should go stiff (hard to turn by hand); the drivers should only get warm.

✓ **Switch ON: both motor shafts go stiff (hard to turn by hand). The drivers get only warm.**

Sensor and laser wires

Sensor through the multiplexer, laser straight in

blue = SDA · green = SCL · orange = 5 V · black = GND



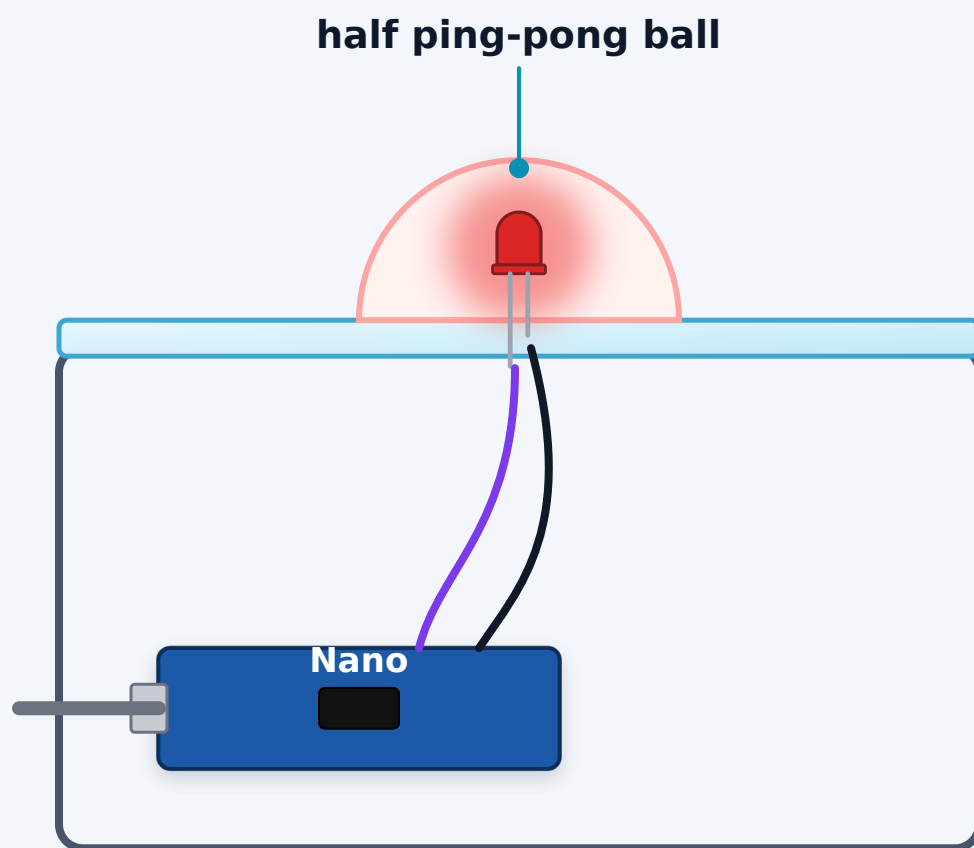
 **Tools / parts for this page:** Multiplexer (TCA9548A), AS5600 sensor, laser module (KY-008), jumper wires.

- 1 Wire the Arduino to the multiplexer: **A4→SDA, A5→SCL**, 5 V→VIN, GND→GND.
- 2 Join the multiplexer's **A0, A1, A2** pins to GND — this fixes its address at 0x70.
- 3 Plug the sensor into the multiplexer's **channel 0** pins (SD0, SC0).
- 4 Wire the sensor itself: VCC→5 V, GND→GND, DIR→GND.
- 5 Wire the laser: **S (signal) → D7**, **—→GND**. Leave its middle pin empty, nothing connects there.
- 6 After uploading the code (card 13), check the Arduino reports **encoders=pan**.

 After uploading the code (card 13) the Arduino says **encoders=pan**. That means the sensor works.

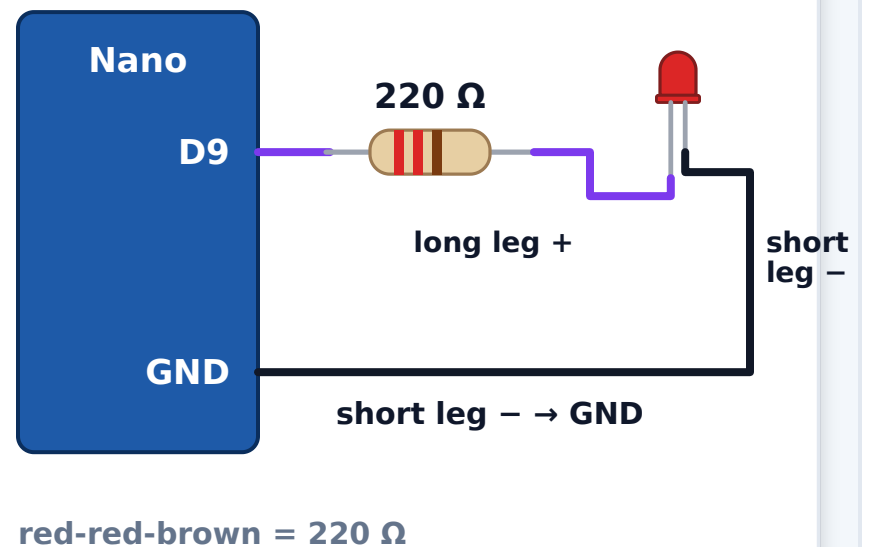
The beacon (the blinking target)

Cut through the middle

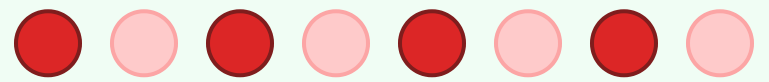


USB cable out of a side hole →
phone charger or power bank

THE CIRCUIT



BLINKS 4 TIMES A SECOND



Upload `tools/rig/beacon_firmware.ino`
to the **SECOND** Nano.

Slow-motion video: count 4 flashes / s.

Tools / parts for this page: Second Arduino Nano, red LED, 220 Ω resistor, ping-pong ball, hot glue, phone charger or power bank.

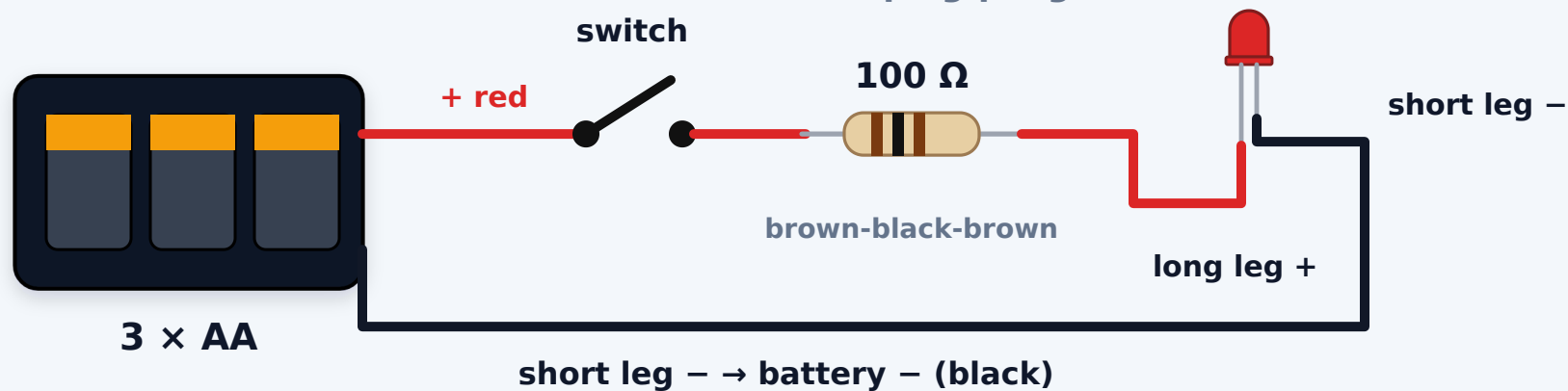
- 1 Wire the LED's **long leg** → 220 Ω resistor → the Nano's **D9** pin.
- 2 Wire the LED's **short leg** straight to the Nano's **GND** pin.
- 3 Upload **beacon_firmware.ino** (a separate, simpler sketch) to this second Nano.
- 4 Cut a ping-pong ball in half. Poke the LED through a hole in the box lid, and glue one half over it as a diffuser.
- 5 Power this Nano from a phone charger or power bank — a separate supply from the rig.
- 6 Check the LED blinks steadily, 4 times a second. No light? Swap the LED's 2 legs around.

✓ LED blinks steadily. No light? The LED is the wrong way round — swap its two wires.

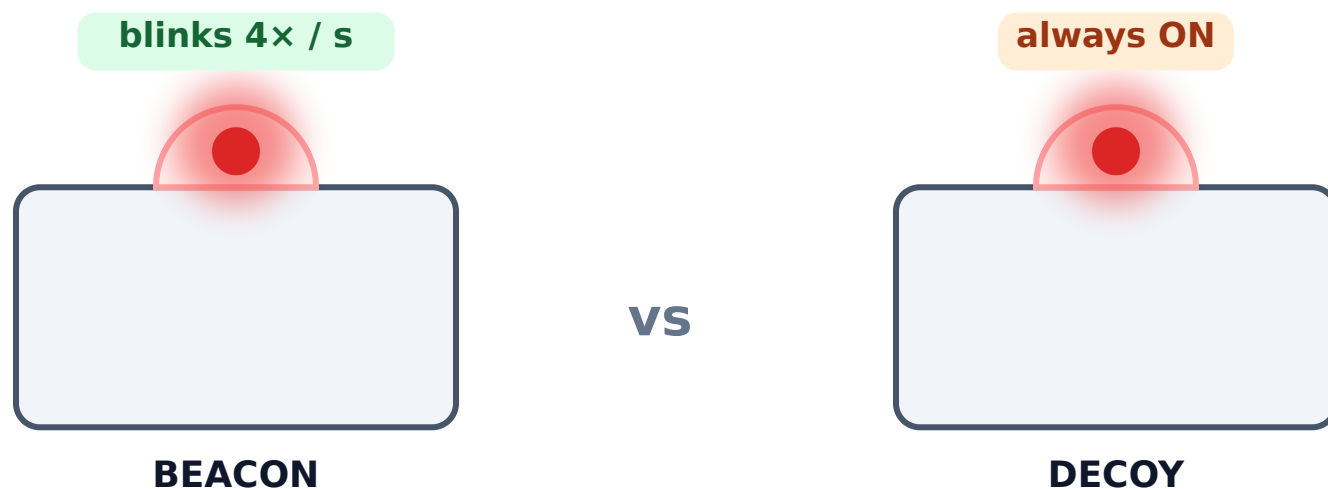
The decoy (the trick light)

The decoy: no Arduino, no code

Same red LED, same box, the other half of the ping-pong ball



MAKE THEM LOOK THE SAME



Tools / parts for this page: 3× AA battery holder, switch, red LED, 100 Ω resistor, same style box + the other ping-pong ball half.

- 1 Wire: battery + → switch → **100 Ω** resistor → LED's **long leg**.
- 2 Wire: LED's **short leg** → battery – (black wire) directly. No Arduino and no code here at all.
- 3 Build the same kind of box, with the **other half** of the ping-pong ball, so it looks exactly like the beacon.
- 4 Flip the switch on — it should glow steady and **never blink**.
- 5 No light? The LED is in backwards — swap its 2 legs around.

✓ Switch on: a steady red light. No light? Flip the LED round.

Put the code in the Arduino

Arduino IDE

- 1 Tools → Manage Libraries → search **AccelStepper** → Install
- 2 File → Open → **tools/rig/rig_firmware_v2.ino**
- 3 Tools → Board: **Arduino Nano** · Processor: **ATmega328P (Old Bootloader)**
- 4 Tools → Port: the Nano → press **UPLOAD** (→)
- 5 Tools → Serial Monitor · **115200 baud** · Newline

```
# ZeroDrift Mk2 ready, encoders=pan
```

TYPE THESE TO TEST (press Enter after each)

P267 T0

bottom motor turns 30°

P0 T0

back to the start

T89

top motor nods 10°

L1 · L0

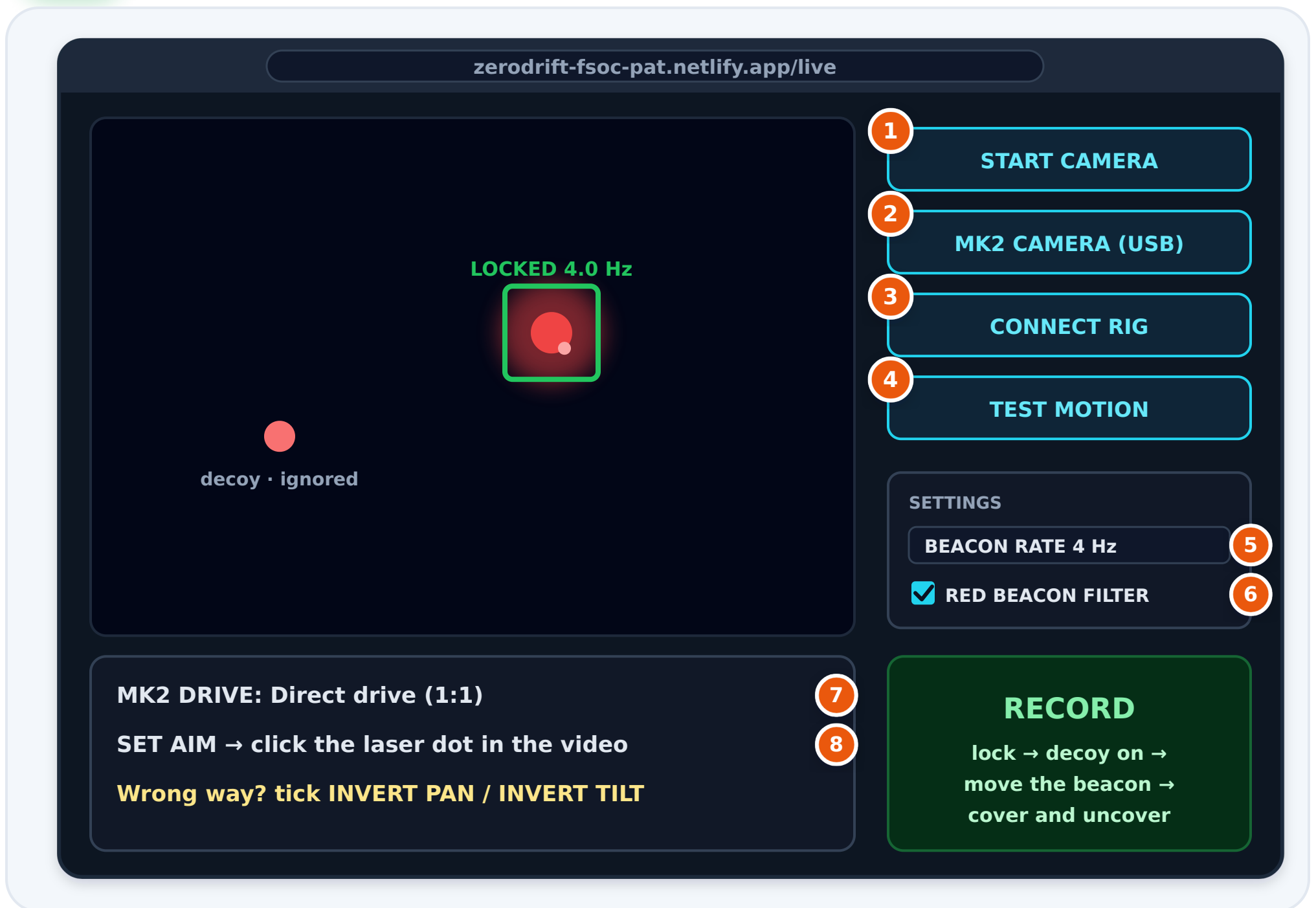
laser on · off

Tools / parts for this page: Laptop, USB cable, Arduino IDE installed.

- 1 Plug the rig's Arduino into the laptop with its USB cable.
- 2 Install the library: Tools → Manage Libraries → search **AccelStepper** → Install.
- 3 Open **tools/rig/rig_firmware_v2.ino** in the IDE, then set the Board to Arduino Nano.
- 4 Set Processor to **ATmega328P (Old Bootloader)** — try the other option if upload fails.
- 5 Set the Port to the Nano, then click **Upload** (→) and wait for "Done uploading".
- 6 Open the Serial Monitor at 115200 baud, type **P267 T0** and press Enter — the bottom motor should turn.
- 7 **Close the Serial Monitor** afterwards — the website can't connect while it stays open.

✓ You see "ZeroDrift Mk2 ready" and the motors move when you type **P267 T0**.

Go live on the website



Tools / parts for this page: Laptop with Chrome, webcam or laptop camera, the rig's Arduino connected by USB.

- 1 Open Chrome and go to the **/live** page on the website.
- 2 Click **START CAMERA**, then pick **MK2 CAMERA (USB)** — or the laptop camera if the webcam hasn't arrived.
- 3 Click **CONNECT RIG**, choose the Arduino's port, then click **TEST MOTION** to confirm it moves.
- 4 Motor moving the wrong way? Tick **INVERT PAN** or **INVERT TILT**.
- 5 Turn the beacon on. In a bright room, tick **RED BEACON FILTER**.
- 6 Click **SET AIM**, then click the laser dot inside the video — this tells the software where it is aiming right now.

✓ It locks in about a second and turns to face the beacon. Now record the demo!